



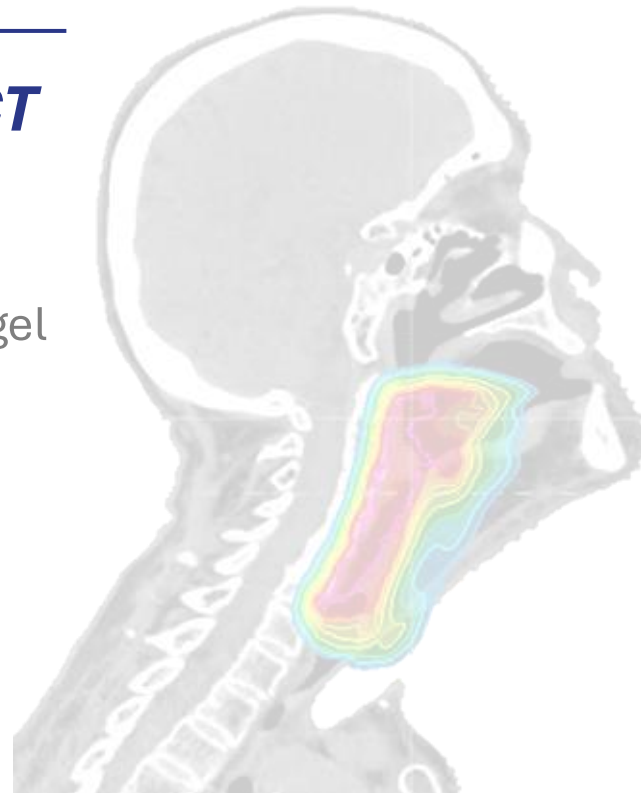
Can we trust synthetic CT algorithms?

Uncertainty-aware evaluation of CBCT to CT synthesis for Adaptive Proton Therapy

Blanca Rodriguez-Gonzalez, Ivan Escobar Corominas, Angel
Torrado-Carvajal, Borja Rodriguez-Vila, Alejandro Mazal,
Norberto Malpica

MIART
Medical Image AI in Radiation Therapy

Thursday, October 1st





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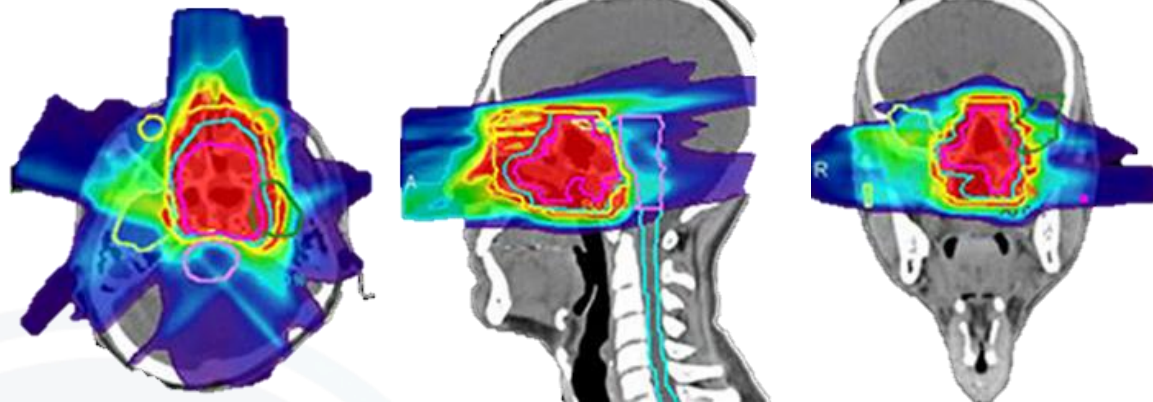
- PhD Student at **PROMISE Research** group at Universidad Rey Juan Carlos (Madrid, Spain)
- Through our collaboration with **Centro de Protonterapia Quirónsalud**, we develop AI methods to support **adaptive proton therapy**.

Authors declare no conflict of interest.

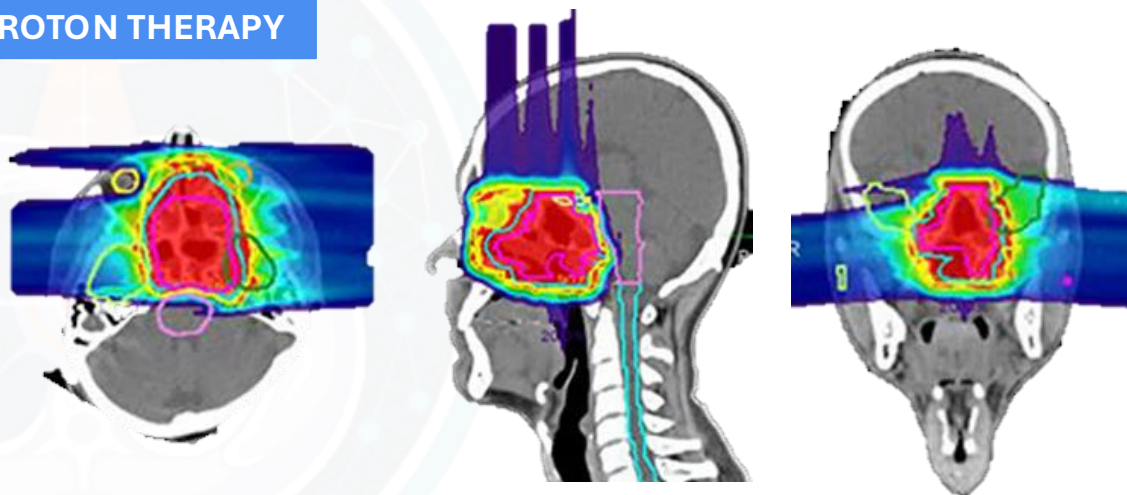
This study has been funded by the **MAGERIT-CM** project (TEC-2024/COM-44), funded by Comunidad de Madrid.

Proton Therapy in a nutshell

PHOTON THERAPY

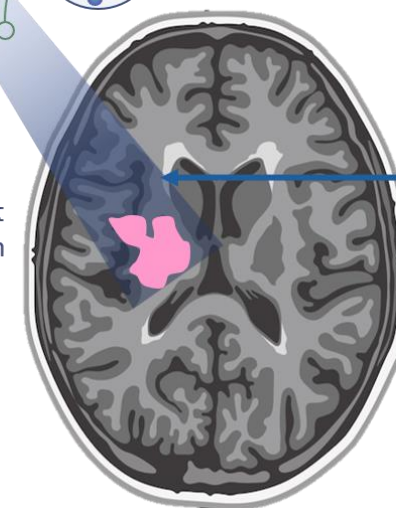


PROTON THERAPY



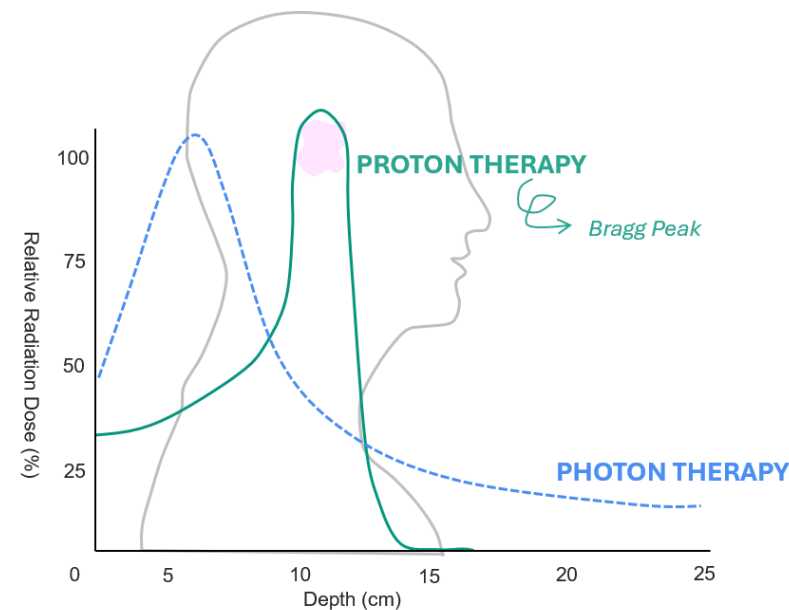
Protons are tiny charged particles

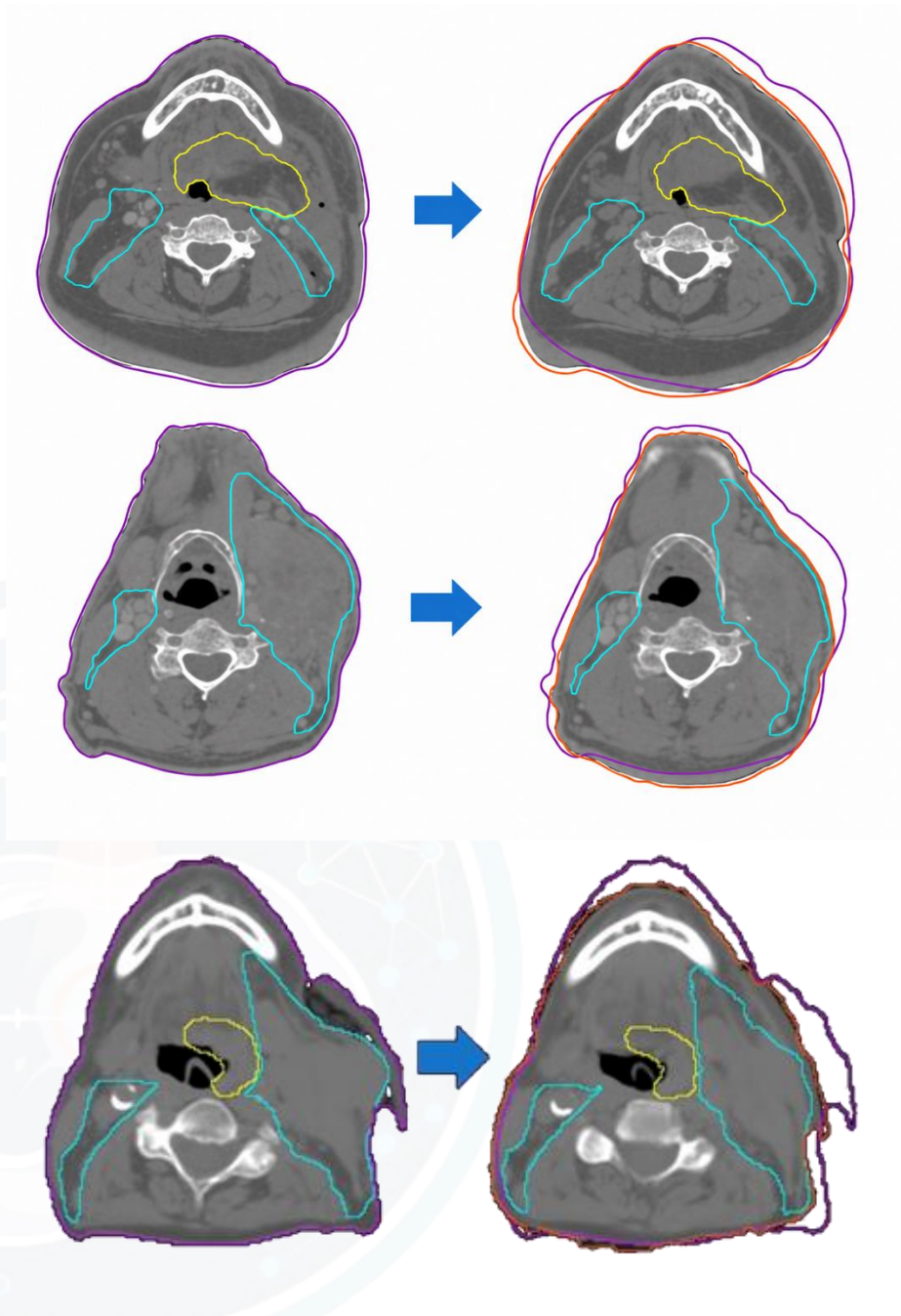
Proton release a sudden burst of energy when they stop ...



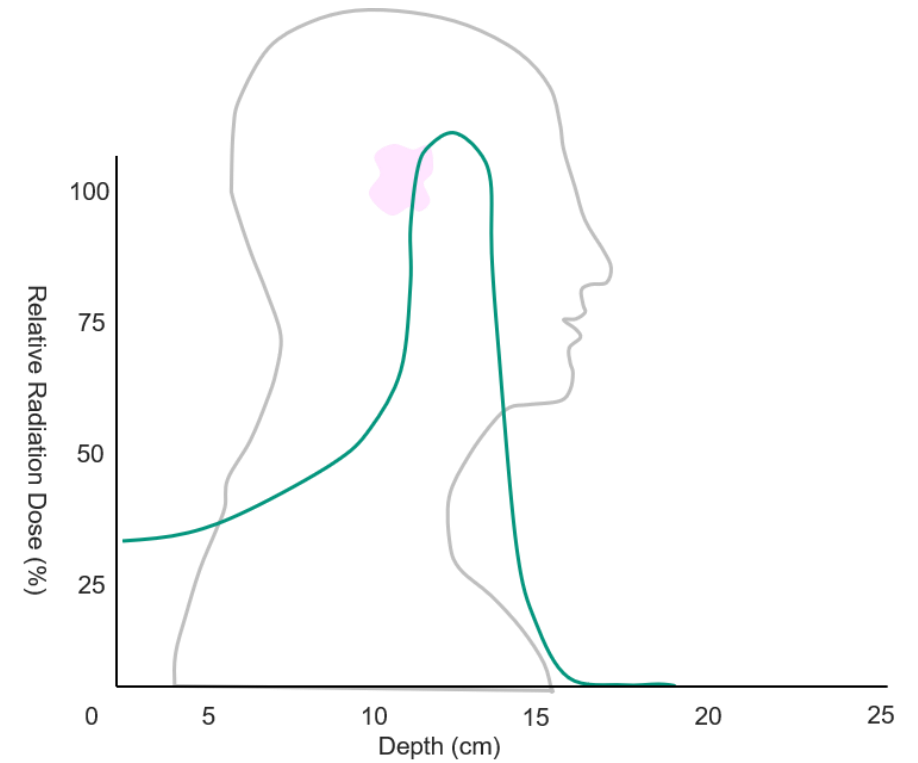
LESS HEALTHY CELLS DAMAGED BY THE BEAM

... which 'stays inside the tumor'

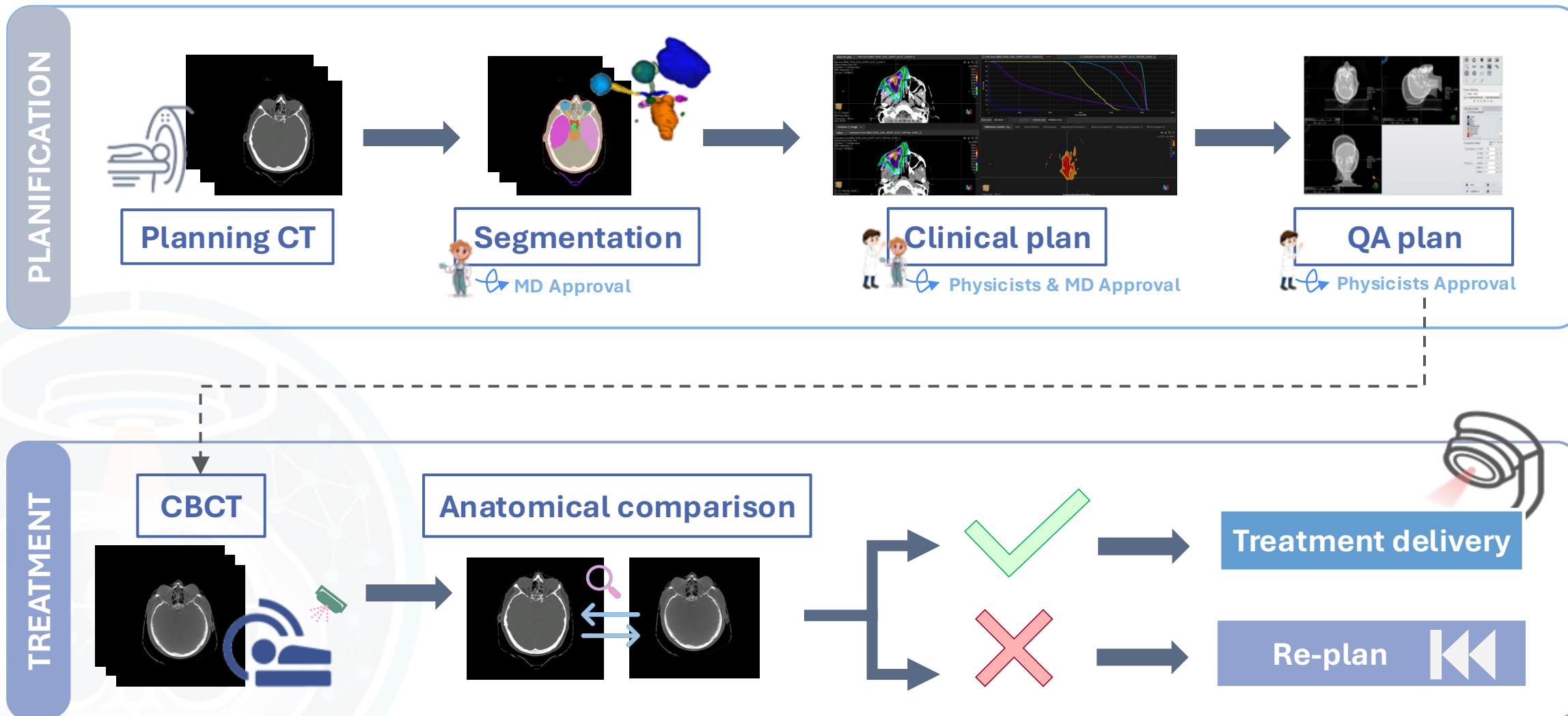




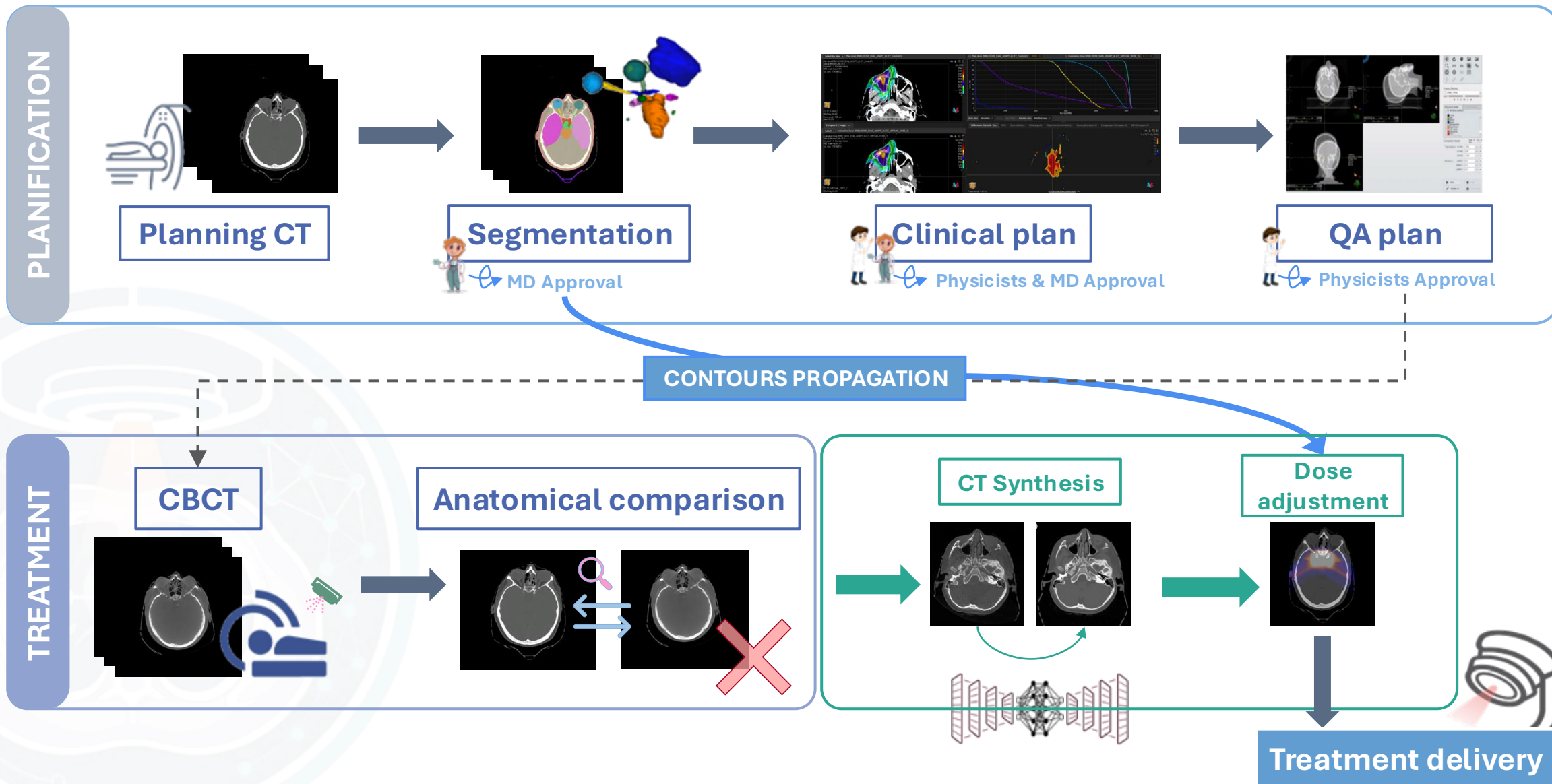
The same physics that makes proton therapy precise also makes it vulnerable to **anatomical changes**.



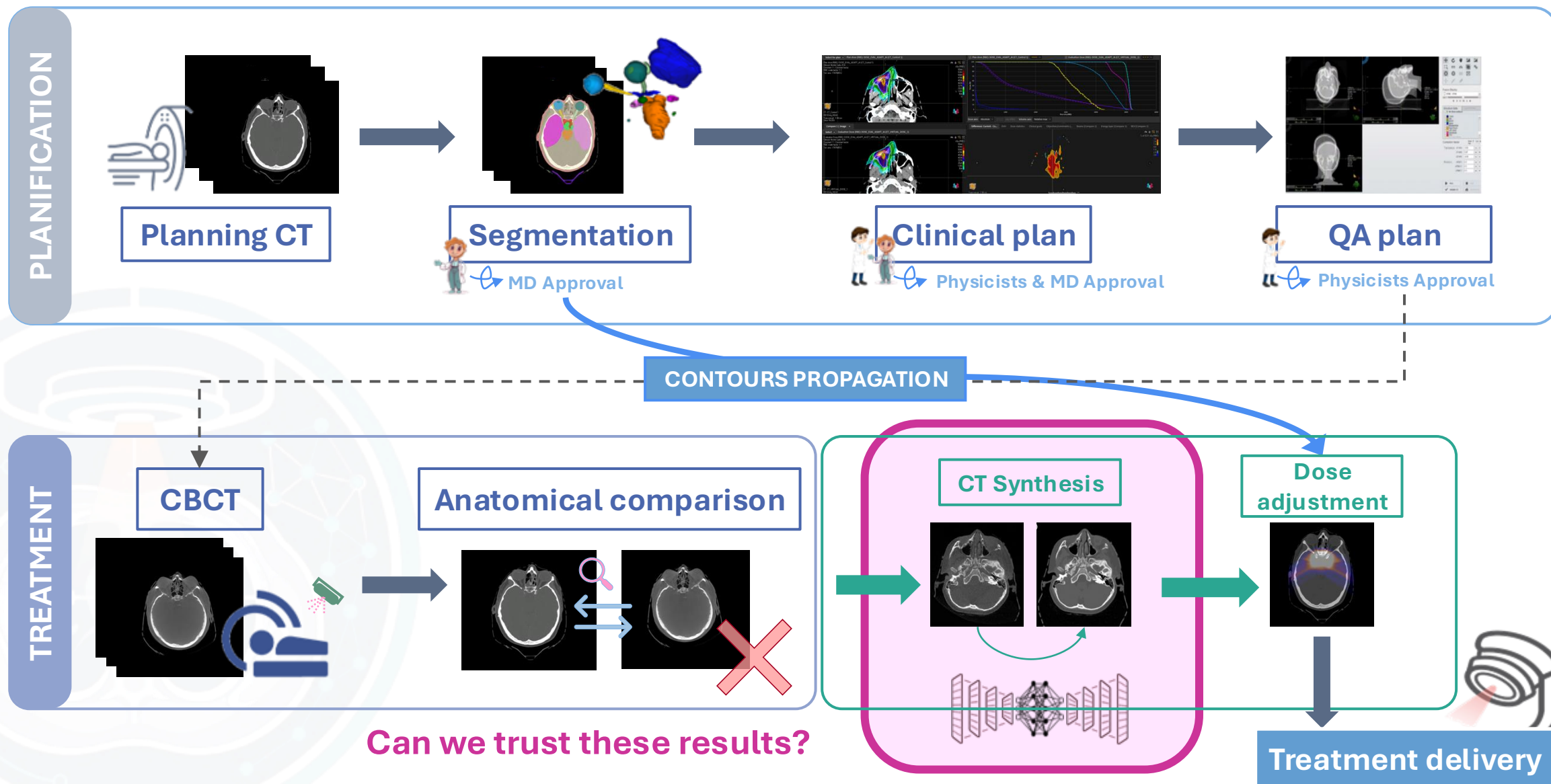
“Online” APT



sCT-based “online” APT

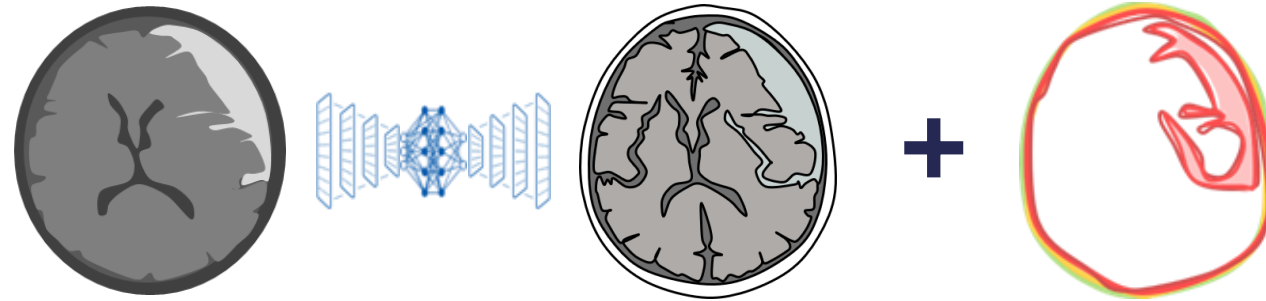


sCT-based “online” APT



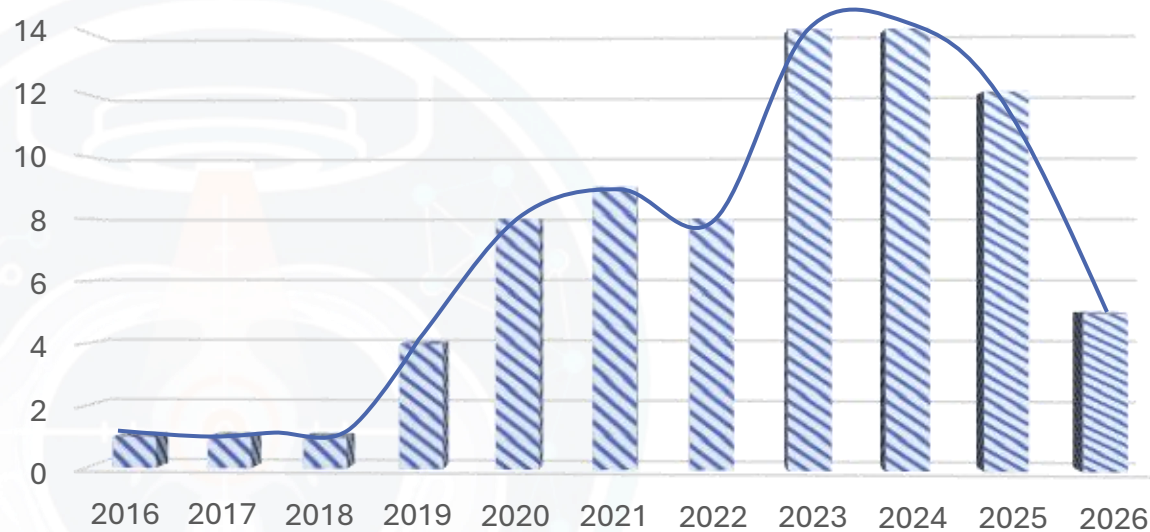
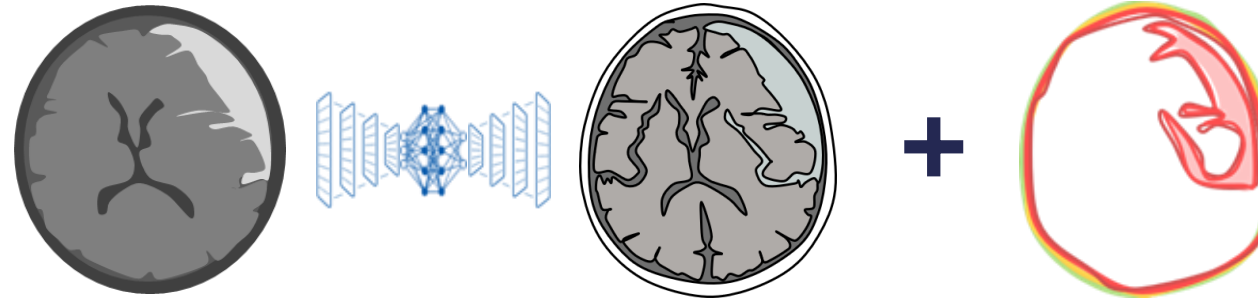
UNCERTAINTY QUANTIFICATION

*Instead of asking only what the model predicts, we want to know **how sure the model is***



UNCERTAINTY QUANTIFICATION

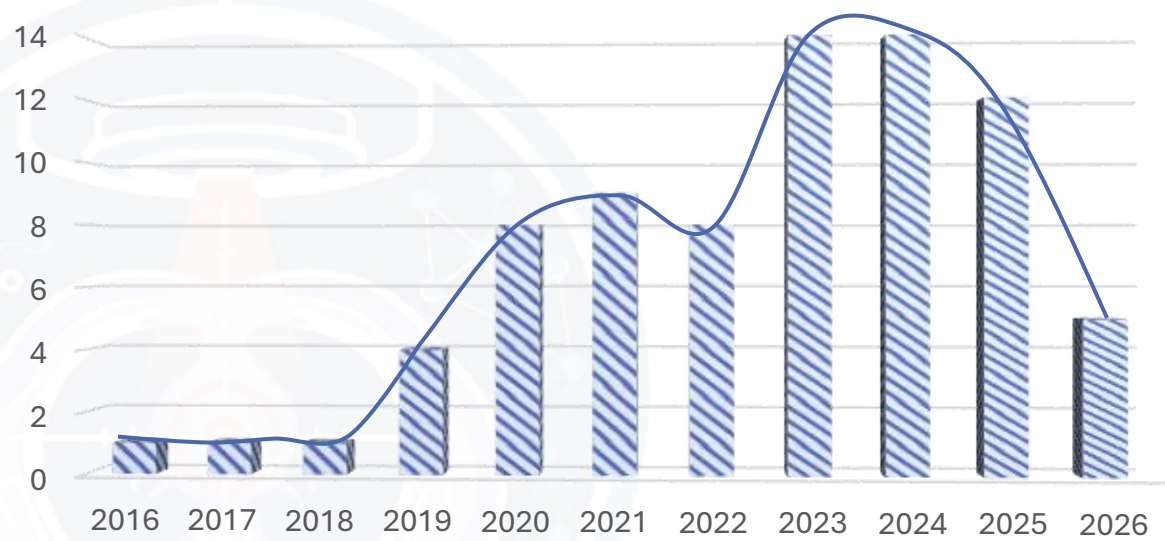
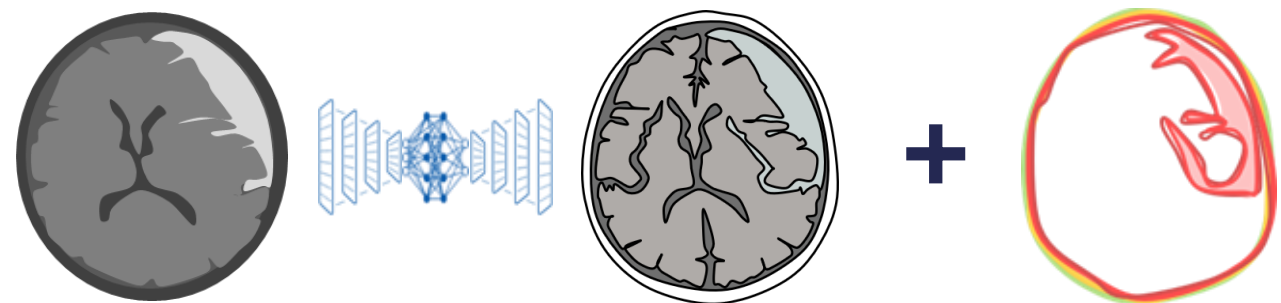
*Instead of asking only what the model predicts, we want to know **how sure the model is***



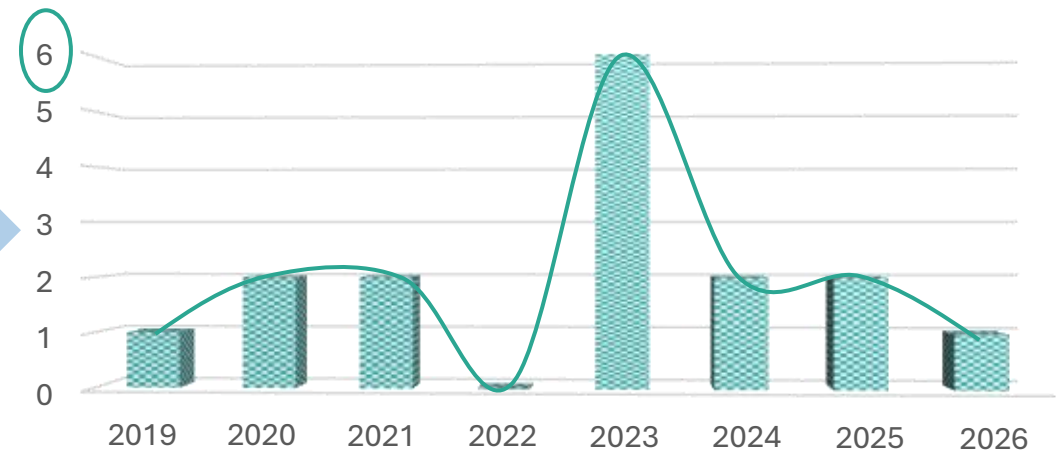
((synthetic CT **OR** CT synthesis **OR** sCT)
AND (proton therapy **OR** adaptive proton therapy))

UNCERTAINTY QUANTIFICATION

Instead of asking only what the model predicts, we want to know **how sure the model is**



((synthetic CT **OR** CT synthesis **OR** sCT)
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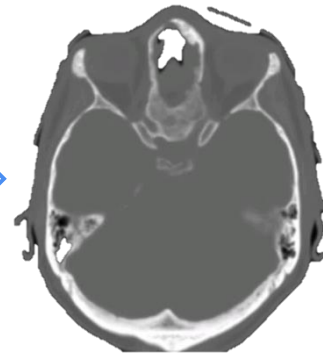
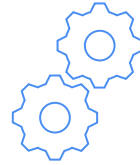


((synthetic CT **OR** CT synthesis **OR** sCT)
AND (proton therapy **OR** adaptive proton therapy))
AND (uncertainty **OR** reliability **OR** confidence **OR** trustworthy)

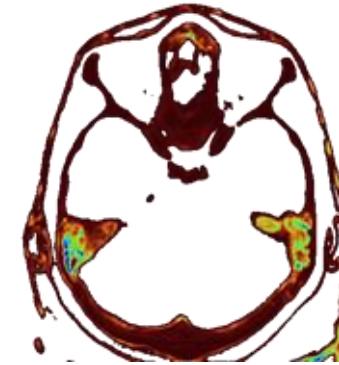
UNCERTAINTY QUANTIFICATION



Daily CBCT



Synthetic CT



UQ Map

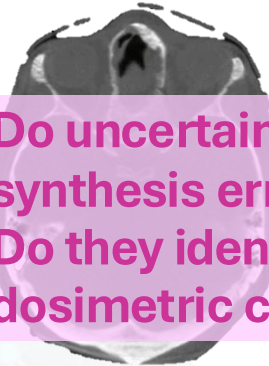


+

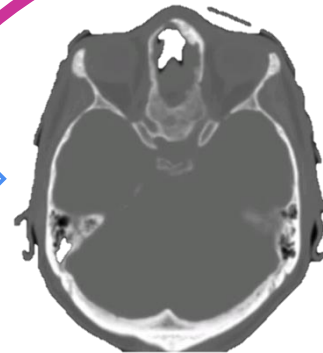


Do uncertainty maps provide useful information?

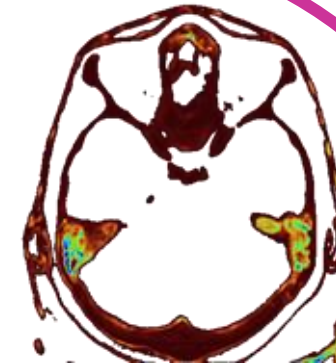
- Do uncertainty maps localize synthesis errors? ➡
- Do they identify regions with dosimetric consequences?



Daily CBCT



Synthetic CT



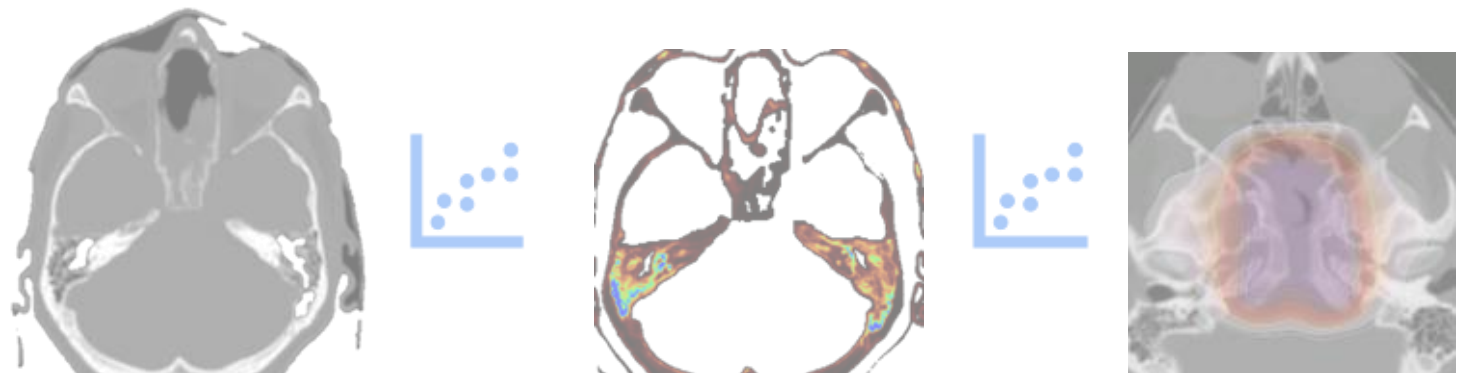
UQ Map

HYPOTHESIS

Meaningful uncertainty estimates should spatially coincide with synthesis errors and highlight regions with clinically relevant dose discrepancies.

OBJECTIVE

To compare different UQ methods for CBCT-to-CT synthesis and assess whether their uncertainty estimates localize image and dose errors.



THE DATA SET

INCLUSION CRITERIA

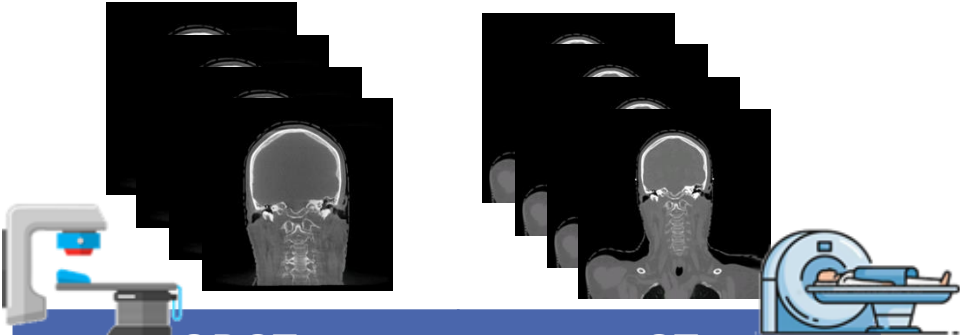
- Head and Neck cancer diagnosis.
- Underwent proton therapy treatment before December 2024.
- Availability of at least one CBCT-CT image pair acquired within a maximum interval of 7 days.

Centro de Protonterapia

Grupo quirónsalud

84 patients

270 pairs CBCT-CT
[2-7] pairs per patient



	CBCT	CT
Manufacturer	IBA	GE
Matrix size	512x512x~257	512x512x~257
Slice thickness	1.250 mm	1.000 mm
Pixel size	0.537 mm	0.977 mm

THE DATA SET: Ensuring Data Quality

PRE-PROCESSING

1 Convert DICOM and RTSTRUCT to NIfTI



2 Registration to CBCT space

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

3 HU standarization on CBCT and CT

Establish HU limits
[-1024,3071]



MIN-MAX Normalization
[0,1]

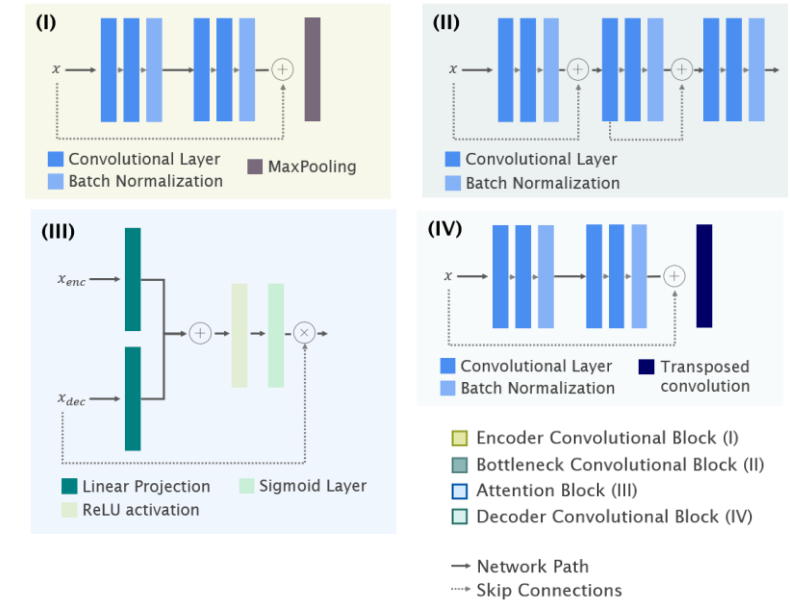
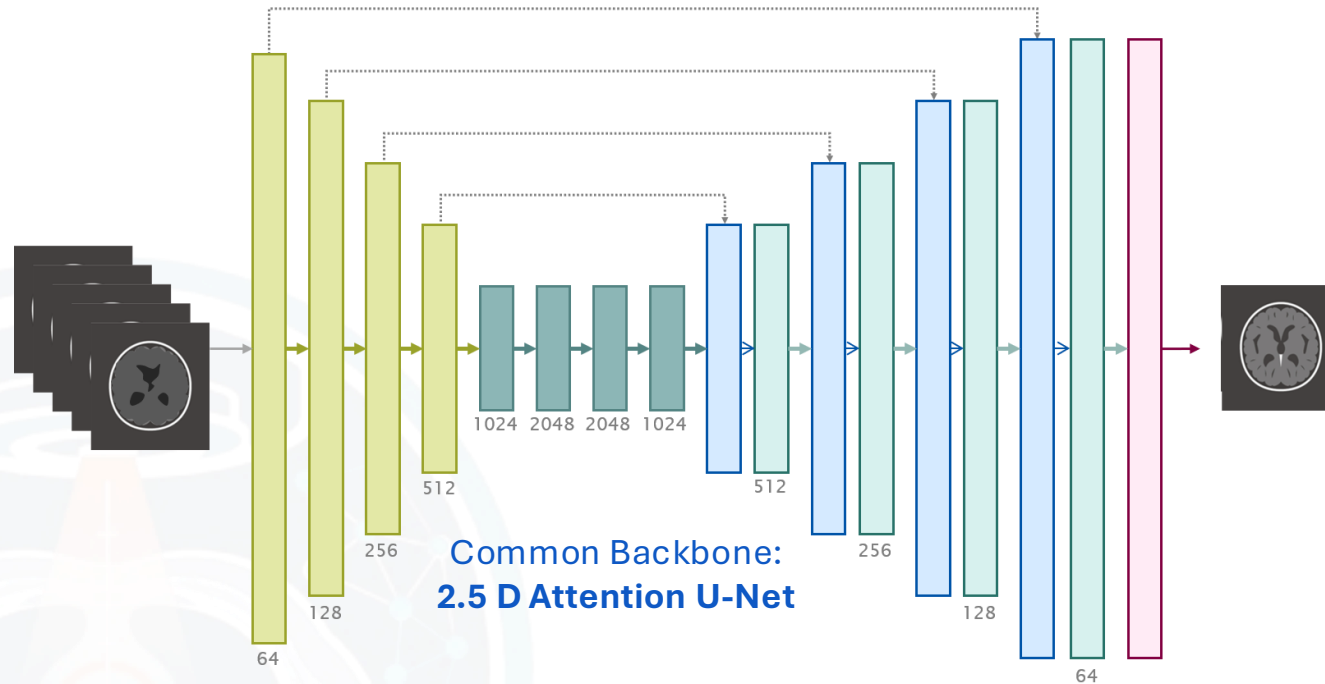
4 Patient-wise train (80%), validation (15%) and test (5%) splits + 5 Fold CV



Although **random** splits were created, generalizability was assessed

HOW? Mean intensity (before step 3) was computed for each CT and CBCT on each set. Statistical tests between mean distributions was assessed. Intensity is not statistically different. Generalizability is ensured.

UQ Methods Strategies



UQ Methods Strategies

Monte Carlo (MC) Dropout



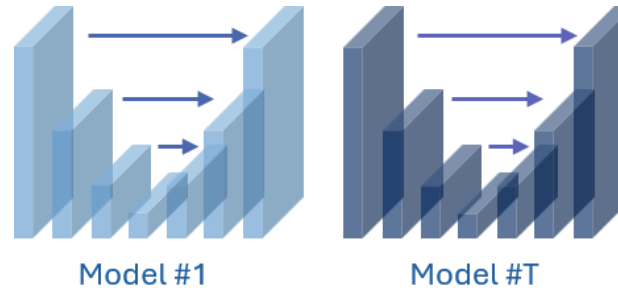
15 Stochastic Passes. **0.25** Dropout rate



sCT: mean prediction
UQ map: variance across predictions

$$\mathcal{L} = \lambda_1 \text{MAE} + \lambda_2 (1 - \text{SSIM}) + \lambda_3 \text{Gradient Loss}$$

Deep Ensembles

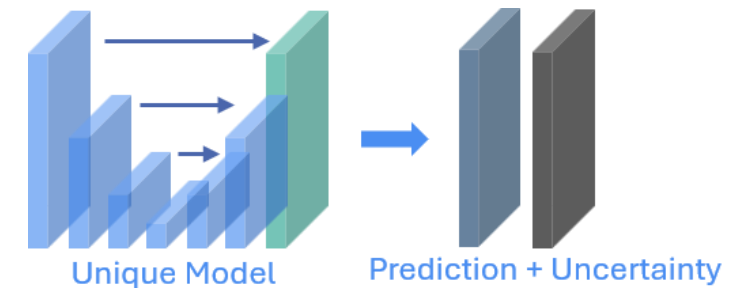


5 Independently train models



sCT: mean prediction
UQ map: variance across predictions

Evidential Deep Learning



$p(y | \mu, \lambda, \alpha, \beta) = \text{NIG}(y | \mu, \lambda, \alpha, \beta)$
with $\lambda, \beta > 0$ and $\alpha > 1$



sCT: μ - predictive mean

UQ map: $\sigma_{\text{alea}}^2 = \frac{\beta}{\alpha - 1} + \sigma_{\text{epis}}^2 = \frac{\beta}{\lambda(\alpha - 1)}$

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \lambda \mathcal{L}_{\text{evidence}}$$

Evaluating UQ Tasks

How accurate is the synthesized CT?

“Traditional” Evaluation Metrics

- MAE [HU]
- PSNR [dB]
- MS-SSIM
- Dice[HU>300]

Dose Impact

- MAE [Gy]
- P95
- Gamma Pass Rate
- CTV ΔD_{95}

How accurately can dose be recalculated?

Uncertainty Calibration Metrics

- UQ-Error Correlation
- UEO
- High error AUROC
- ECE
- NLL

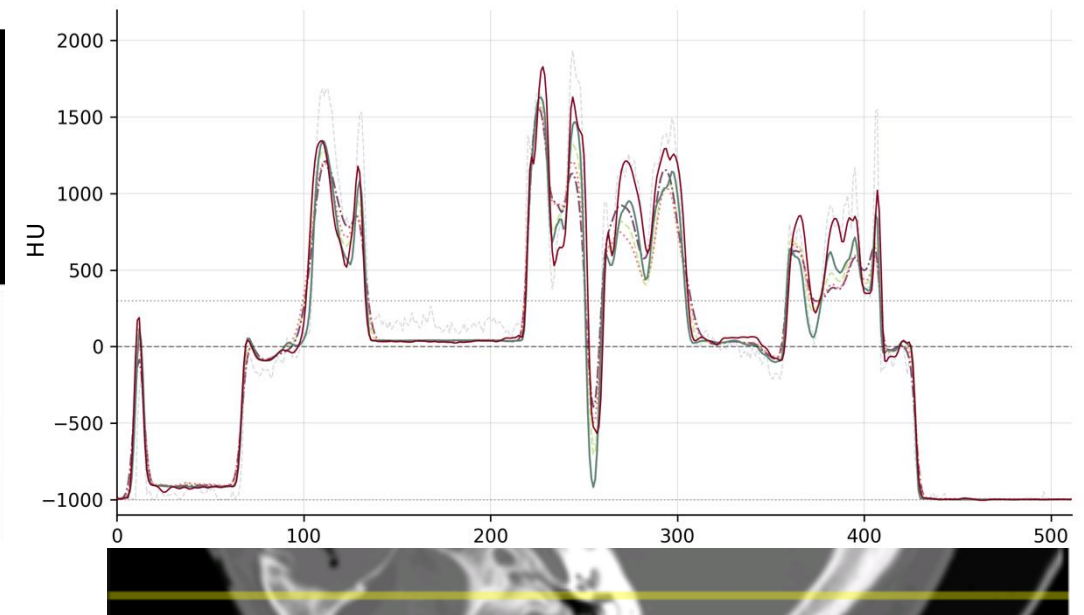
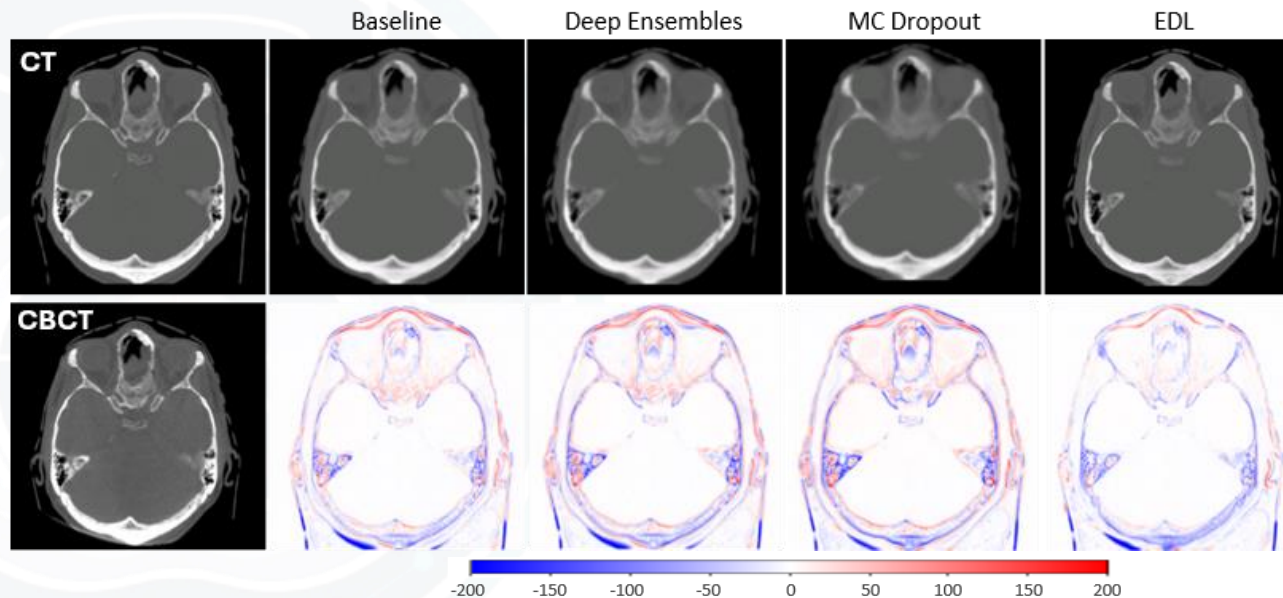
Does uncertainty reflect prediction reliability?

Traditional Synthesis Evaluation Metrics

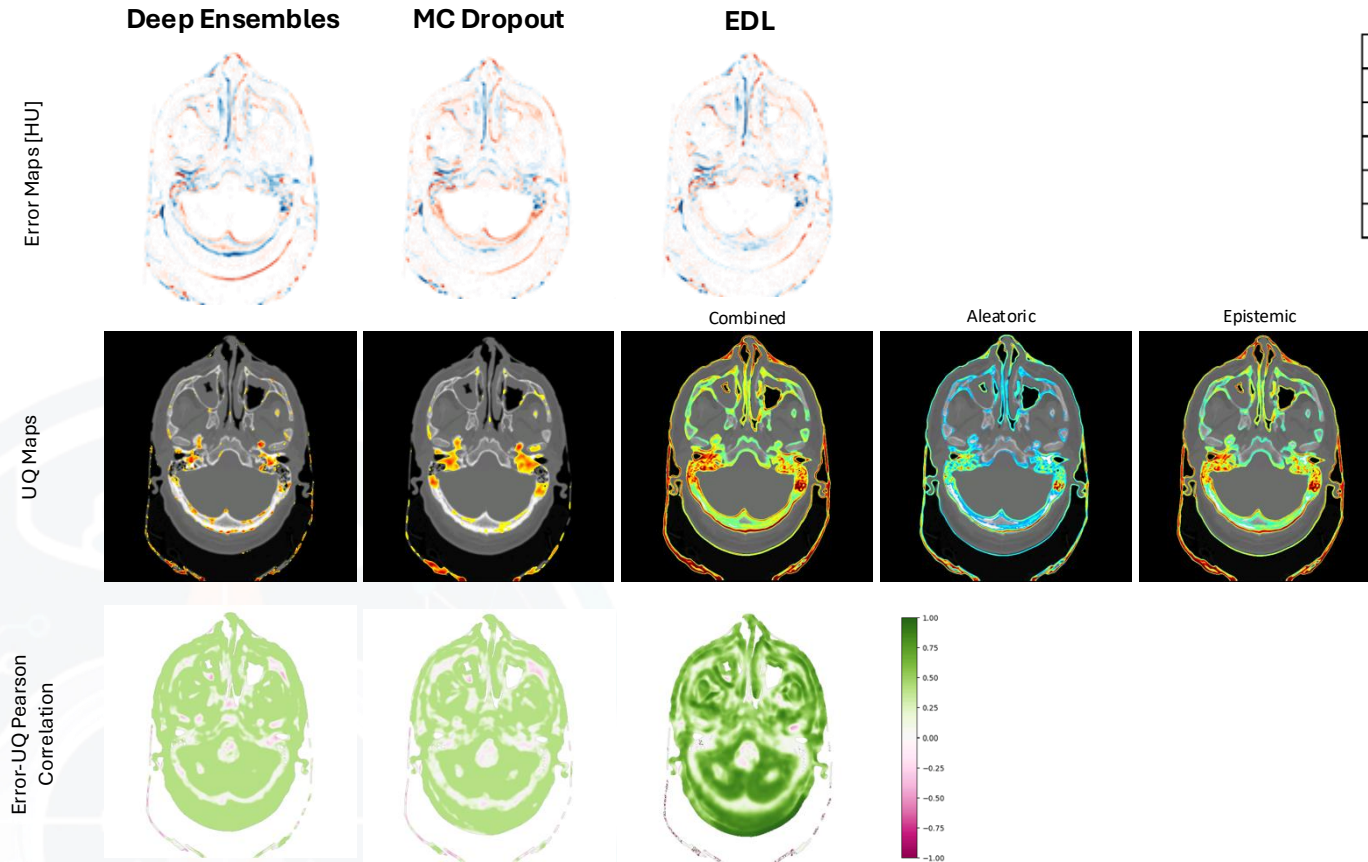
Introducing uncertainty estimation does not compromise image quality. Uncertainty estimates came at no cost.

		Baseline	MC Dropout	Deep Ensemble	EDL
sCT	MAE [HU]	33.754 ± 7.317	36.611 ± 5.322	36.625 ± 7.132	31.361 ± 5.849^{B,M,D}
	PSNR [dB]	30.301 ± 1.570	29.622 ± 1.101	29.892 ± 1.367	31.995 ± 1.538^M
	MS-SSIM	0.959 ± 0.011	0.952 ± 0.010	0.955 ± 0.011	0.958 ± 0.012
	Dice _{HU>300}	0.908 ± 0.018	0.880 ± 0.014	0.891 ± 0.015	0.911 ± 0.024^{M,D}
CTV	MAE [HU]	43.735 ± 34.567	61.796 ± 43.494	52.272 ± 42.537	37.425 ± 30.853^{B,M,D}
	PSNR [dB]	37.624 ± 8.096	33.688 ± 8.048	36.558 ± 8.702	38.654 ± 8.768^{B,M,D}
	SSIM	0.949 ± 0.040	0.910 ± 0.066	0.924 ± 0.061	0.965 ± 0.026^{B,M,D}

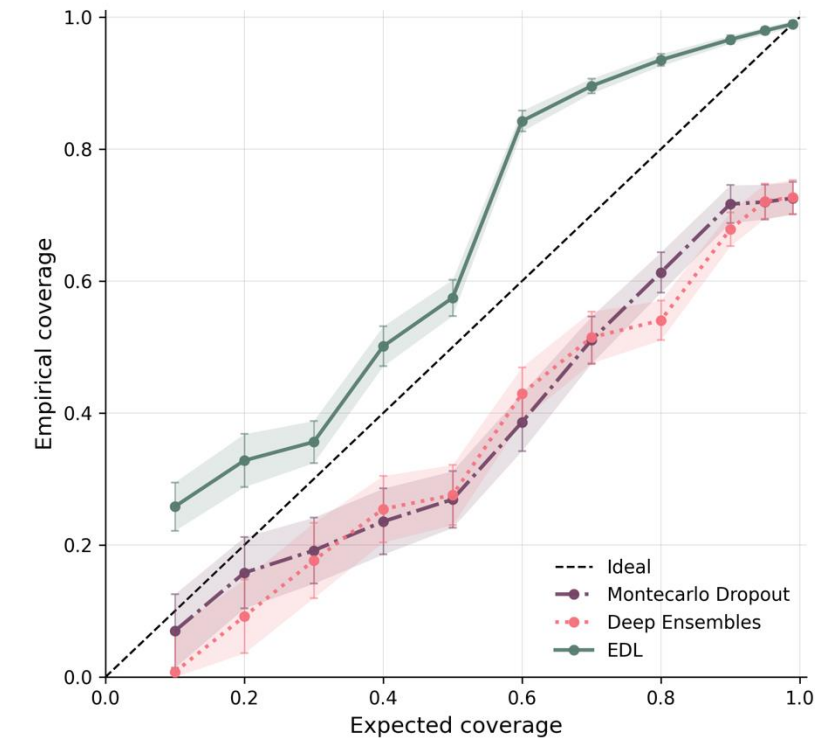
EDL loss function provide a beneficial regularization in terms of synthesis quality.



Can uncertainty identify where synthesis fails??



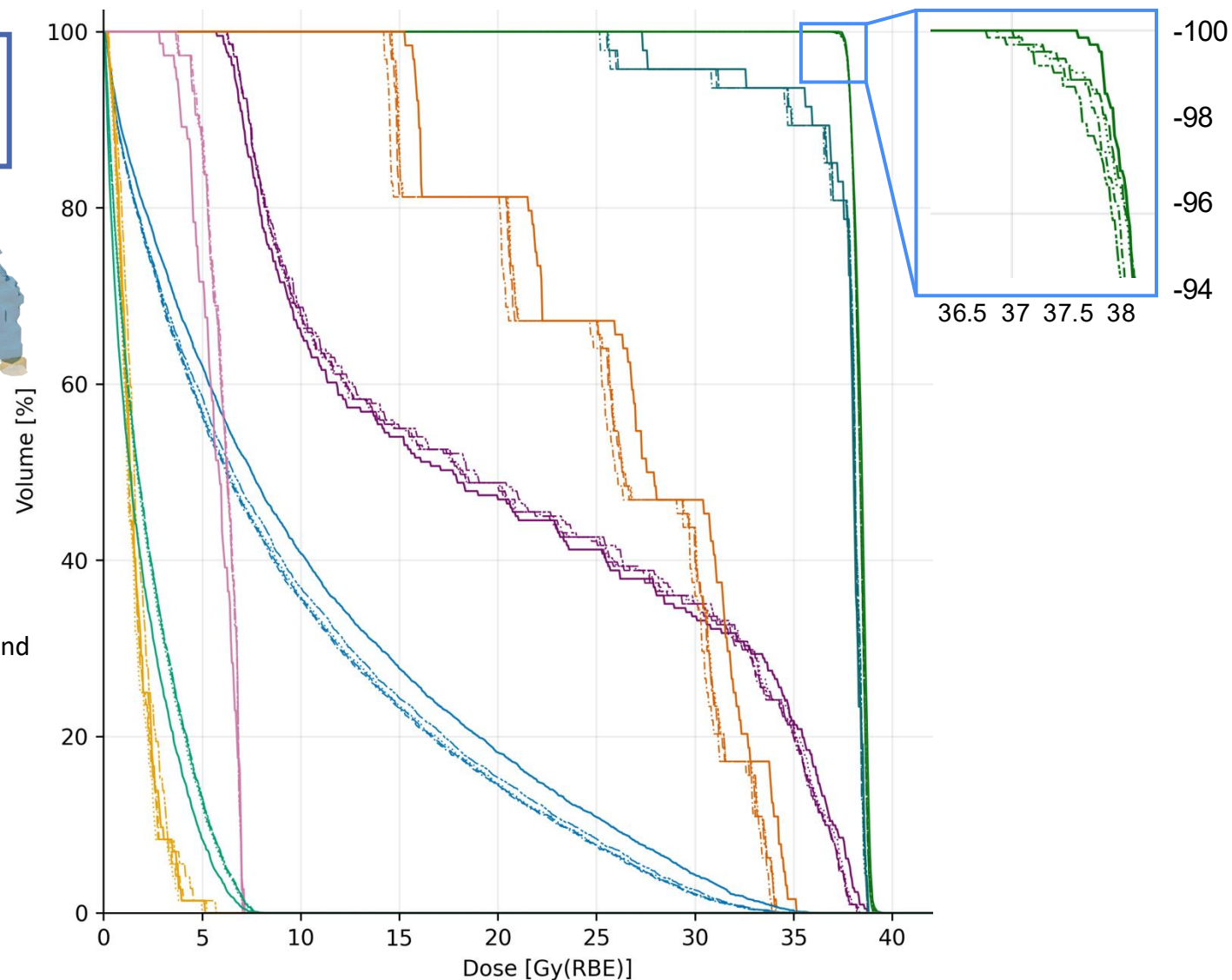
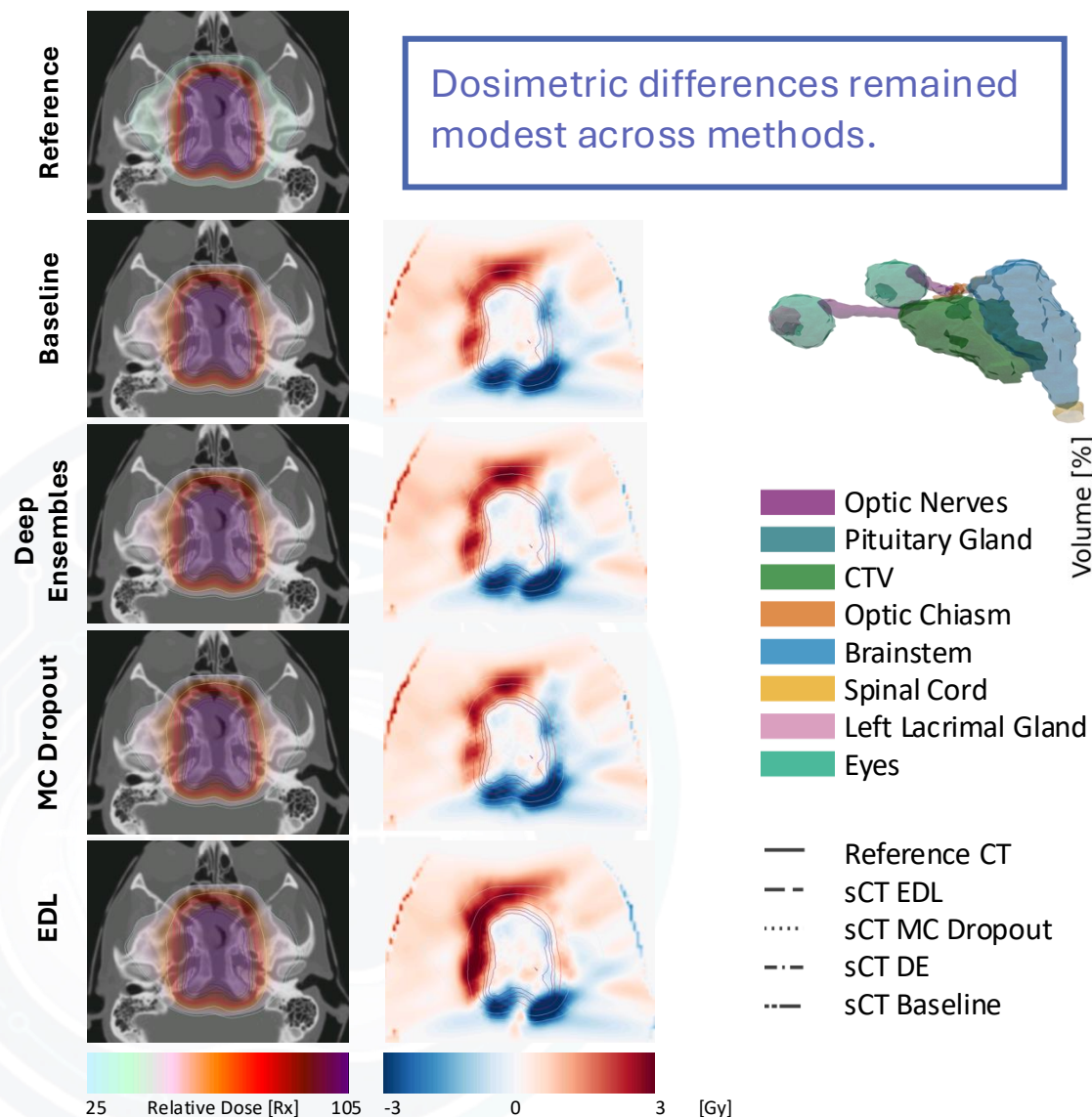
	MC Dropout	Deep Ensemble	EDL
Spearman ($ e , \sigma$)	0.713 ± 0.031	0.704 ± 0.057	$0.774 \pm 0.016^{M,D}$
UEO Dice	0.718 ± 0.025	0.705 ± 0.030	$0.887 \pm 0.034^{M,D}$
High-error AUROC	0.846 ± 0.032	0.826 ± 0.020	0.874 ± 0.017^D
ECE	0.331 ± 0.016	0.322 ± 0.035	$0.136 \pm 0.022^{M,D}$
NLL	11.160 ± 5.683	9.995 ± 4.940	$7.857 \pm 2.761^{M,D}$



Regions with high uncertainty consistently coincide with larger HU errors.

Do synthetic CTs preserve dosimetric accuracy?

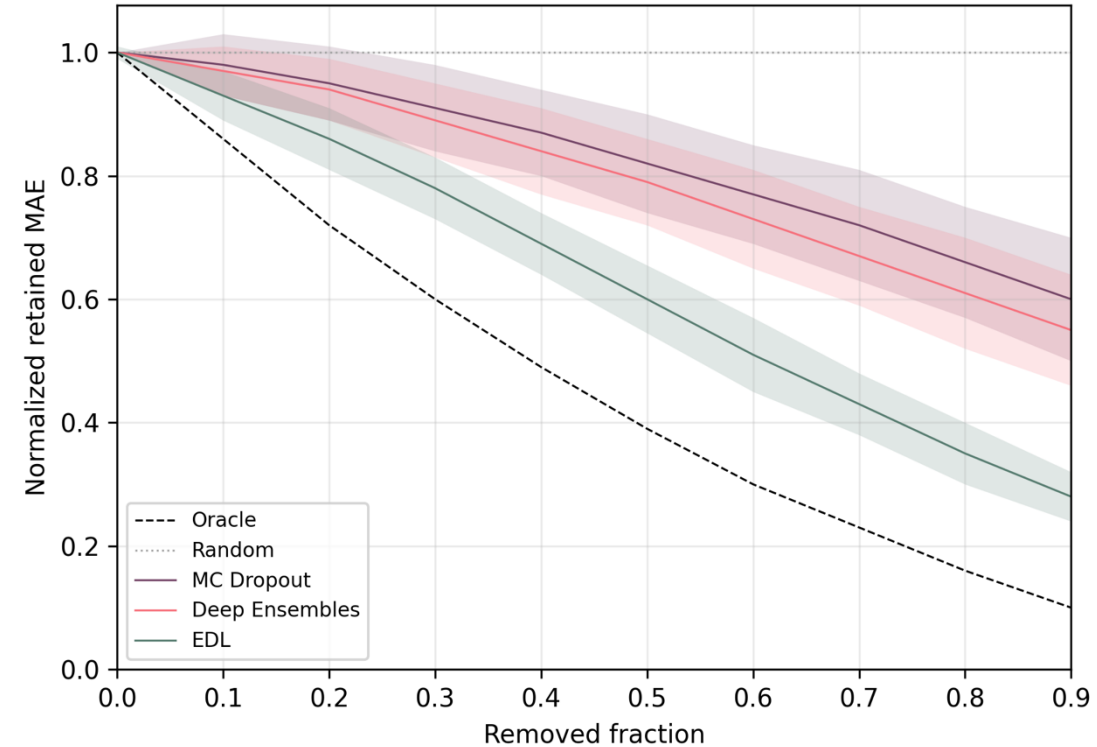
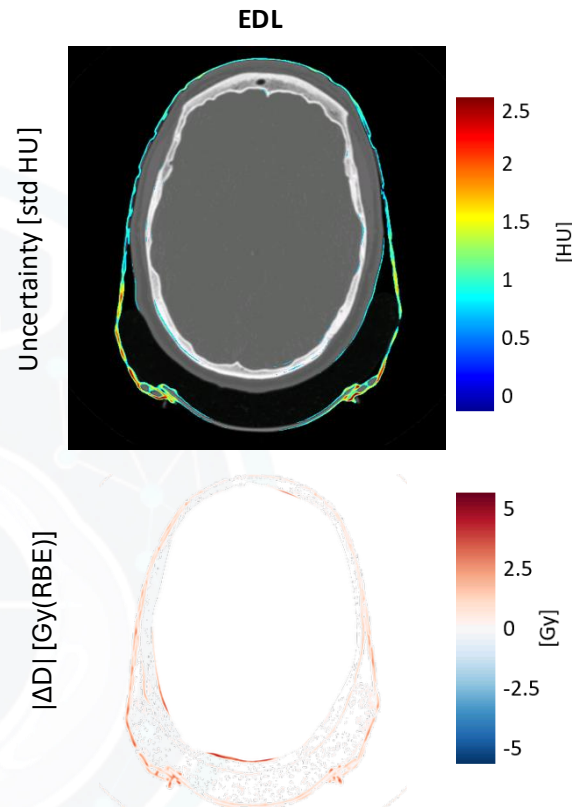
Dosimetric differences remained modest across methods.



What about dosimetric reliability?

	MC Dropout	Deep Ensemble	EDL
Spearman ($ e , \sigma$)	0.698 ± 0.021	0.717 ± 0.047	$0.769 \pm 0.018^{M,D}$
UEO Dice	0.616 ± 0.156	0.605 ± 0.269	$0.669 \pm 0.120^{M,D}$
High-error (10%) AUROC	0.649 ± 0.019	0.693 ± 0.022	0.724 ± 0.011

High-uncertainty regions concentrated larger dose discrepancies, with EDL shows the strongest dose-error association.



LIMITATIONS OF THE STUDY



Despite improved calibration, uncertainty estimates should not be interpreted as perfectly calibrated probabilities of error.



Other sources of proton therapy uncertainty were not considered (i.e., SPR, anatomical deformations, range uncertainty)



Interpretation of EDL uncertainty: Although EDL achieved the best results; evidential estimates depend on the adopted prior and regularization strategy.

UNCERTAINTY $\neq$ ERROR

High uncertainty often indicates increased error risk, but low uncertainty does not guarantee a correct prediction.

CORRELATION $\neq$ CAUSATION

We evaluate uncertainty through its association with HU and dose errors, but uncertainty is not directly propagated through the dose calculation process.

CONCLUSIONS: TAKE HOME MESSAGES

HYPOTHESIS

Meaningful uncertainty estimates should spatially coincide with synthesis errors and highlight regions with clinically relevant dose discrepancies.

Uncertainty maps were informative of both synthesis and dose errors, with Evidential Deep Learning achieving the strongest error-uncertainty agreement and the best calibration.

What we evaluated

- ★ Three different UQ methods integrated in a 2.5D Attention U-Net.
- ★ sCT quality
- ★ Uncertainty-error localization
- ★ Uncertainty Calibration
- ★ Dose-error Localization

What we found

- ★ UQ methods maintained competitive synthesis performance.
- ★ EDL improved image quality metrics while providing uncertainty estimates.
- ★ Uncertainty consistently co-localized with HU errors.
- ★ High-uncertainty regions concentrated larger dose discrepancies.
- ★ EDL achieved the best calibration and uncertainty-error agreement.

WHY?

APT requires not only accurate sCTs, but also reliable estimates of when and where predictions may fail. Uncertainty should be considered a complementary quality-assurance signal rather than just an additional image.



Can we trust synthetic CT algorithms?

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